

Strongly Exponential Sensitive Monotone Bounds in VP via Eulerian Conditioning

Ying Xie
Kennesaw State University
yxie2@kennesaw.edu

Abstract

We present a strongly exponential sensitive monotone circuit lower bound for an explicit family of arborescence polynomials in uniform VP. If N is the actual number of variables, P_H is the arborescence polynomial, and F_H is the product of the outgoing-choice group sums, then $\text{monSIZE}(F_H \pm \epsilon P_H) \geq 2^{bN}$ for fixed $a, b > 0$ and $2^{-aN} \leq \epsilon \leq 1/2$. The circuits may use arbitrary nonnegative real constants and unrestricted fan-out. The construction conditions independent local permutations of an expander's oriented edges to form one Eulerian cycle. The BEST theorem and two spectral trace identities bound the conditioning cost, while permutation exposure supplies a concentration estimate strong enough to absorb that cost. A fixed clique blow-up realizes an inner-product gadget and yields one distribution with exponentially small discrepancy for every balanced partition of the variable groups. The resulting coefficient argument charges shared gates. As consequences, a uniform VP family with coefficients one and two has a sharp multiplicative approximation threshold of $\sqrt{2}$, even for approximation on the whole cube without a degree restriction, and strongly exponential exact monotone counting complexity on Boolean inputs.

1 Introduction

Monotone arithmetic circuits compute polynomials using addition, multiplication, and nonnegative constants. General arithmetic circuits can additionally exploit cancellation. Sensitive lower bounds ask whether a small coefficient perturbation of an easy polynomial can already force a large monotone circuit. The perturbation parameter is part of the statement: hardness at a fixed perturbation does not by itself imply hardness for exponentially small perturbations.

Hrubeš introduced a systematic study of this phenomenon [6]. Chattopadhyay, Datta, and Mukhopadhyay related sensitive monotone lower bounds to communication complexity [1]. Chattopadhyay, Datta, Ghosal, and Mukhopadhyay [2] proved ordinary $2^{\Omega(N)}$ monotone lower bounds for sparse spanning-tree polynomials in VP, and sensitive $2^{\Omega(\sqrt{N})}$ bounds for a complete-graph family in VP. Here and throughout this comparison, N counts actual input variables. Their Section 7 asks for strongly exponential sensitive bounds. Chattopadhyay, Ghosal, and Mukhopadhyay subsequently obtained such sensitive bounds for a graph-inner-product family in VNP [3]; that result does not supply a VP upper bound.

1.1 Main result

For a graph H with a distinguished root, assign one variable $x_{z,w}$ to each arc from a nonroot vertex z to a neighbor w . Let F_H be the product, over nonroot vertices, of the sums of their outgoing

variables. Let P_H sum precisely those full-group monomials whose chosen arcs lead every vertex to the root. The matrix-tree determinant computes P_H by polynomial-size general arithmetic circuits.

Theorem 1.1 (Asymptotic form). *There are positive constants a, b and explicit polynomial families (F_N, P_N) on exactly N variables, for every sufficiently large N , such that F_N has an $O(N)$ -size monotone circuit, P_N is in uniform VP, and*

$$\text{monSIZE}(F_N \pm \epsilon P_N) \geq 2^{bN} \quad \text{whenever} \quad 2^{-aN} \leq \epsilon \leq \frac{1}{2}.$$

Each target has the full support of F_N and coefficients 1 and $1 \pm \epsilon$. The lower bound allows arbitrary nonnegative real constants and unrestricted sharing.

The core family is given by a fixed clique blow-up of a simple constant-degree expander. Theorem 2.1 provides numerical constants and a finite cutoff. Section 5 supplies a uniform explicit family and the padding argument that proves Theorem 1.1. The constants are conservative; degree and blow-up factors are fixed before the family size tends to infinity.

The point is the simultaneous combination of VP membership, strongly exponential size in actual variables, and sensitivity down to an exponentially small perturbation. The ordinary strongly exponential VP separation is already known [2]. No claim about Boolean circuit lower bounds or about separating general arithmetic complexity classes follows from the present theorem.

1.2 Proof mechanism

The proof uses the discrepancy-to-coefficient framework of [1, 2]. Its central construction is a distribution on rooted neighbor maps of a sparse host which has exponentially small rectangle discrepancy under every balanced partition of the outgoing variable groups. A single distribution must work for all partitions; the sampled layout itself need not be simultaneously good for all of them.

Let B be a simple d -regular base with $v = d|V(B)|$ oriented edge states. Independent local permutations define a cycle cover of these states. Condition the cover to consist of one cycle. The BEST theorem [4] expresses the probability of this event through the spanning-tree count of B . The identities for the first and second spectral traces show that the reciprocal conditioning probability is at most

$$v \exp(O(v/d^2)).$$

Keeping the signed first trace, rather than replacing it by an absolute-value estimate, is important for this bound.

A fixed number of clones per state realizes a two-track parity gadget along this Eulerian cycle using only edges of the host. For each fixed balanced partition, expansion guarantees a positive expected density of stages whose first bit is held entirely on one side and whose second bit is held entirely on the other. Revealing permutation images individually gives a tail bound $\exp(-\Omega(v))$. At a sufficiently large fixed degree this tail absorbs the conditioning cost. The retained stages give a large inner product, whose elementary discrepancy estimate yields the desired universal bound.

Finally, a nonnegative full-group output forces productive gates to be set-multilinear. Removing one balanced gate at all its uses charges at most one product to that gate. This gives the circuit-size bound even with unrestricted fan-out.

1.3 Consequences and related models

The polynomial $R_H = 2F_H - P_H$ has coefficients one and two on every full-group monomial. Section 6 proves an exact approximation threshold: any fixed symmetric multiplicative factor below $\sqrt{2}$ requires $2^{\Omega(N)}$ monotone size, whereas $\sqrt{2}F_H$ attains factor $\sqrt{2}$ with linear size. This holds coefficientwise and uniformly on $[0, 1]^N$, without a degree bound on the competing circuit. The passage from the cube to coefficients uses least-homogeneous-component extraction, a standard monotone tool [7, 8].

Section 7 extends the size bound to exact arithmetic counting on the whole Boolean cube. We give a full-group normalization with explicit cost $(2g + 1)s + g$, where g is the number of groups. Removing repeated consistent indicators has classical precedents in sum-product networks [11, 10]; the normalization principle is not claimed as a new contribution. The full-cube equality hypothesis is essential to this application.

The main argument occupies Sections 2–5. Appendix A gives the direct BEST counting argument, Appendix B reduces the host degree, and Appendix C extends the construction to any fixed-degree simple expander family with a fixed positive spectral gap after a suitable fixed clique blow-up.

2 Polynomials and circuit model

All arithmetic circuits are directed acyclic graphs with binary $+$ and $\times$ gates. Size counts arithmetic gates, not input occurrences. Monotone circuits use nonnegative real constants, and fan-out is unrestricted. General circuits may use negative rational constants. A uniform VP family has polynomially many variables, polynomial degree, and a polynomial-time constructible family of polynomial-size general arithmetic circuits. When a real ϵ is not specified by a uniform rational sequence, it is treated as a supplied scalar parameter.

A polynomial is *full-group set-multilinear* for a partition $X_1, \dots, X_g$ when each monomial uses exactly one variable from each group, with exponent one. The notation $[m]Q$ denotes a coefficient. For a graph B , the clique blow-up $B[K_t]$ replaces each vertex by a t -clique and each edge by a complete bipartite graph between the corresponding cliques. All logarithms in probability estimates are natural unless their base is displayed.

Fix integers $d \geq 734$, $L \geq 32$. Let B be a connected simple d -regular graph on n vertices with

$$\lambda_2(A_B/d) \leq \rho := 2/27. \quad (2.1)$$

Write $v = dn$. Let $H = B[K_{dL}]$, a fixed clique blow-up of the base graph. Its degree is $D_H = (d + 1)dL - 1$. Choose any fixed root r . Give every directed arc out of a nonroot vertex a different variable, so the actual input count is

$$N = (Lv - 1)D_H. \quad (2.2)$$

Define

$$F_H = \prod_{z \neq r} \sum_{w \in N_H(z)} x_{z,w}, \quad P_H = \sum_{\nu: \text{all vertices reach } r} \prod_{z \neq r} x_{z,\nu(z)}.$$

The sum is over all choices of one neighbor at each nonroot vertex. The rooted out-Laplacian determinant equals P_H , so F_H, P_H have uniform polynomial-size general arithmetic circuits when B is explicit.

Theorem 2.1 (Finite sensitive bound). *Put $K = LD_H$, $c = 1/7680000$, $a = c/(2K)$, and $b = c/(3K)$. For every B satisfying (2.1) with $v \geq 2^{32}$,*

$$\text{monSIZE}(F_H \pm \epsilon P_H) \geq 2^{bN} \quad (2^{-aN} \leq \epsilon \leq 1/2). \quad (2.3)$$

The constants a, b depend on the fixed d, L . The exponent is not uniform in growing degree or clone count.

The competing monotone circuits have binary addition and multiplication, arbitrary nonnegative real constants, and unrestricted fan-out. Size counts arithmetic gates. Their intermediate gates are not assumed set-multilinear. The output coefficients are all positive: they are 1 and $1 \pm \epsilon$, and their support is that of the easy product F_H .

The communication domain is the full product $\mathcal{X}_H = \prod_{z \neq r} N_H(z)$. A partition assigns whole outgoing-choice groups to the two players and is balanced if each side contains between $(Lv - 1)/3$ and $2(Lv - 1)/3$ groups. For a distribution Δ , discrepancy is the largest absolute signed mass

$$\left| \sum_{\nu \in \mathcal{U} \times \mathcal{V}} \Delta(\nu)(2g_H(\nu) - 1) \right|$$

over rectangles for that partition, where g_H is the arborescence indicator. There is no promise on allowable neighbor maps.

2.1 Determinant upper bound

Let $h = |V(H)|$. Omit the row and column of the root and use the row vector

$$\sum_{w \in N_H(z)} x_{z,w}(e_z - e_w), \quad e_r := 0$$

at each $z \neq r$. Row multilinearity of the determinant shows that the coefficient of $m_\nu = \prod_{z \neq r} x_{z,\nu(z)}$ is the determinant with rows $e_z - e_{\nu(z)}$. If every vertex reaches the root, a common ordering of rows and columns by an acyclic order makes this matrix triangular with diagonal one. Its determinant is one. Otherwise choose a nonroot directed cycle and put a vector equal to one on that cycle's entire basin and zero elsewhere. Its difference across every selected outgoing arc is zero. It is a nonzero kernel vector, so the determinant is zero. Thus the determinant is exactly P_H .

The determinant has uniform polynomial-size general arithmetic circuits over rational constants. The construction of H is uniform as described below, and F_H is a product of group sums. Consequently $F_H \pm \epsilon P_H$ has a polynomial-size circuit with ϵ as a parameter or a supplied constant. For an unambiguously uniform family over the rationals one may fix $\epsilon = 1/2$, or set $\epsilon_N = 2^{-\lceil aN/2 \rceil}$, which lies in the proved interval for sufficiently large N and has $O(N)$ -bit rational representation. No polynomial-size set-multilinear circuit or polynomial-size formula upper bound is being asserted for P_H .

Parameter	Meaning
n	Number of base vertices
d	Fixed base degree
$v = dn$	Number of oriented edge states
L	Fixed number of clones per state
$h = Lv$	Number of host vertices
$D_H = (d + 1)dL - 1$	Host degree
$N = (h - 1)D_H$	Actual variable count

3 Eulerian distribution and parity embedding

3.1 Local permutations and the conditioning probability

Let S be the oriented edges of B , so $|S| = v$. Independently at each base vertex w , choose a uniform permutation σ_w of its neighbors. Set

$$\pi(u, w) = (w, \sigma_w(u)). \quad (3.1)$$

This is a permutation of S . Immediate reversals are allowed. Let E be the event that π is a single cycle.

Write $t_o(B)$ for the number of spanning trees of B , oriented toward a fixed base vertex o . The BEST theorem [4] gives exactly

$$\Pr[E] = \frac{t_o(B)((d-1)!)^n}{(d!)^n} = \frac{t_o(B)}{d^n}. \quad (3.2)$$

A single cyclic transition configuration corresponds to exactly one Eulerian tour starting with any prescribed directed edge; hence there is no extra factor d , n , or v in the numerator of (3.2). A direct counting proof and an efficient sampler appear in Appendix A.

Let $\theta_1 = 1, \theta_2, \dots, \theta_n$ be the normalized adjacency eigenvalues. The matrix-tree formula yields

$$\Pr[E] = \frac{1}{v} \prod_{i=2}^n (1 - \theta_i). \quad (3.3)$$

Simplicity and regularity give the exact identities

$$\sum_{i=2}^n \theta_i = -1, \quad \sum_{i=2}^n \theta_i^2 = n/d - 1 = v/d^2 - 1. \quad (3.4)$$

For every $0 \leq \rho < 1$ and $-1 \leq x \leq \rho$,

$$\log(1-x) \geq -x - \frac{x^2}{2(1-\rho)}. \quad (3.5)$$

Indeed, if $f(x) = \log(1-x) + x + x^2/[2(1-\rho)]$, then

$$f'(x) = \frac{x(\rho-x)}{(1-\rho)(1-x)}.$$

Thus f decreases to $f(0) = 0$ on $[-1, 0]$ and increases on $[0, \rho]$. This also covers a bipartite eigenvalue -1 . Using (3.4)–(3.5) in (3.3), and $1/[2(1-\rho)] = 27/50$, gives

$$\Pr[E] \geq \frac{1}{v} \exp\left(-\frac{27v}{50d^2}\right). \quad (3.6)$$

The linear trace term in (3.4) is essential: replacing it by the absolute sum of eigenvalues would give a much more expensive conditioning bound.

3.2 A direct parity gadget

Index the states by their tail and a neighbor label, $S = V(B) \times [d]$. Divide each dL -clone base cluster of H into d labeled state clusters I_s of size L . If $s = (u, w)$ and $\pi(s) = (w, z)$, the two tails u, w are adjacent in B . Thus every original transition has every needed physical intercluster edge in H , including reversals. No routing graph is used.

Fix a state s_0 whose cluster contains the root r . Choose four distinct uniform labeled clones

$$a_{s,0}, a_{s,1}, b_{s,0}, b_{s,1}$$

in every state cluster; exclude r in its own cluster. Choices in different clusters are independent. Sample π from the original local-permutation distribution conditioned on E , independently of these clone choices.

For each state $u \neq s_0$, sample independent uniform bits x_u, y_u , and write $w = \pi(u)$. For each track $\ell \in \{0, 1\}$, set

$$a_{u,\ell} \mapsto \begin{cases} a_{w,\ell}, & x_u = 0, \\ b_{w,\ell}, & x_u = 1, \end{cases} \quad b_{w,\ell} \mapsto a_{w,\ell \oplus y_u}. \quad (3.7)$$

The first two destinations are different labeled clones, so both bit choices are different valid outgoing arcs. Set $c_0 = a_{s_0,0}$, $c_1 = a_{s_0,1}$, and

$$c_0 \mapsto a_{\pi(s_0),0}, \quad c_1 \mapsto r. \quad (3.8)$$

The two b -vertices in $I_{\pi(s_0)}$ point to the same-index a -vertices there. Every remaining nonroot clone in I_s points to $a_{s,0}$, using its clique edge.

For every admissible layout and every assignment of the auxiliary bits, the following deterministic identity holds:

$$g_H(\nu) = \sum_{u \neq s_0} x_u y_u \pmod{2}. \quad (3.9)$$

Proof of the parity identity. Fix an admissible layout and all the auxiliary bits. Since the conditioning event E makes π a single cycle, enumerate the states as

$$s_i = \pi^i(s_0) \quad (0 \leq i \leq v-1), \quad s_v = s_0.$$

Here $v \geq 2$, as the underlying base graph is simple and has positive degree. We follow the active stages in the order $s_1, \dots, s_{v-1}$; the transition out of s_0 is instead the special connector and root exit in (3.8).

The outgoing map is well defined. Every a -vertex outside I_{s_0} receives exactly one rule from (3.7), and the two a -vertices in I_{s_0} receive the rules in (3.8). Because π is a permutation, every b -vertex outside I_{s_1} has a unique predecessor state and receives exactly one rule from (3.7). The two b -vertices in I_{s_1} have the separately specified, fixed rules. Every other nonroot vertex is a filler vertex. Thus each nonroot vertex has exactly one outgoing arc, and the root has none. The intercluster arcs follow adjacent tails in B ; all other arcs lie in a cluster clique and have distinct endpoints. In particular, the rules neither conflict nor use a loop.

One stage and the two paths. Put $t_i = x_{s_i} y_{s_i}$ for $1 \leq i \leq v-1$, and let $P = t_1 \oplus \dots \oplus t_{v-1}$. For either incoming rail ℓ , the path from $a_{s_i,\ell}$ to the next a -vertex is exactly one of the following:

$$\begin{aligned} x_{s_i} = 0 & & : & \quad a_{s_i,\ell} \mapsto a_{s_{i+1},\ell} \quad \text{for either value of } y_{s_i}, \\ x_{s_i} = 1, y_{s_i} = 0 & : & \quad a_{s_i,\ell} \mapsto b_{s_{i+1},\ell} \mapsto a_{s_{i+1},\ell}, \\ x_{s_i} = 1, y_{s_i} = 1 & : & \quad a_{s_i,\ell} \mapsto b_{s_{i+1},\ell} \mapsto a_{s_{i+1},\ell \oplus 1}. \end{aligned}$$

Consequently each stage induces the permutation $\ell \mapsto \ell \oplus t_i$ of the two rails. Induction on j shows that the path beginning at $a_{s_1, \ell}$ reaches $a_{s_{j+1}, \ell \oplus t_1 \oplus \dots \oplus t_j}$ after j active stages. In particular, the two paths, stopped at their final a -vertex in I_{s_0} , have endpoints

$$a_{s_1, 0} \rightsquigarrow cP, \quad a_{s_1, 1} \rightsquigarrow c1 \oplus P.$$

These are vertex-disjoint simple paths. Indeed, the two rail indices stay distinct at every state, the intermediate b -vertices are distinct from all a -vertices, and the state order $s_1, \dots, s_v$ contains no repetition. Together the paths contain every a -vertex. Neither contains r .

Odd parity: $P = 1$. The path starting on rail zero ends at c_1 and exits to r . The path starting on rail one ends at c_0 , whose connector leads to the start of the first path. Thus the full forward trajectories are

$$\begin{aligned} a_{s_1, 0} \rightsquigarrow c_1 &\longmapsto r, \\ a_{s_1, 1} \rightsquigarrow c_0 &\longmapsto a_{s_1, 0} \rightsquigarrow c_1 \longmapsto r. \end{aligned}$$

This accounts also for the trajectory from c_0 : its return to I_{s_1} does not close a cycle, since the next active traversal ends at c_1 . Every a -vertex therefore reaches the root.

Even parity: $P = 0$. Now the path starting on rail zero ends at c_0 , so the connector closes the simple directed cycle

$$a_{s_1, 0} \rightsquigarrow c_0 \longmapsto a_{s_1, 0}.$$

This cycle does not contain r . The disjoint path starting on rail one ends at c_1 and then at r . In particular, the outgoing map fails the requirement that every nonroot vertex reach r .

Vertices outside the two paths. Every unused b -vertex points directly to an a -vertex in its own state cluster. This includes the two exceptional b -vertices in I_{s_1} , whose rules are fixed, and the b -vertices bypassed at stages with $x_{s_i} = 0$. Every filler vertex likewise points directly to an a -vertex. Such vertices can create no additional directed cycle. Hence in the odd case all nonroot vertices reach r ; in the even case there is exactly one nonroot directed cycle. By the definition of g_H , its value is respectively one and zero, which proves (3.9).

The argument holds for every admissible layout and every bit assignment, independently of their probability. It uses no spectral estimate, concentration bound, or finite enumeration. $\square$

There are $k = v - 1$ stages. Each x_u affects only the two outgoing choices at $a_{u, 0}, a_{u, 1}$, and each y_u affects only those at $b_{\pi(u), 0}, b_{\pi(u), 1}$.

Let Δ denote the pushforward of this random experiment to the full product of neighbor maps. Thus a layout consists of the conditioned permutation and the clone labels, and the remaining randomness consists of the auxiliary bits x_u, y_u . These auxiliary bits are distinct from the formal arc variables $x_{z, w}$. The distribution is fixed before a communication partition is chosen.

4 Universal discrepancy and the size lower bound

4.1 Expansion and monochromatic pairs

Fix any balanced partition of all nonroot physical vertices. For analysis assign the root either color. Let a_s be the Alice fraction in I_s , and $\mu = v^{-1} \sum_s a_s$. For $Lv \geq 100$, $\mu \in [33/100, 67/100]$.

The one-point transition matrix for the unconditioned cover is $P((u, w), (w, z)) = 1/d$. It is doubly stochastic. The symmetric matrix $Q = (P + P^\top)/2$ has image in the functions $a(u) + b(w)$.

For each base eigenvalue θ , its matrix on the associated two lifted functions is

$$\frac{1}{2} \begin{pmatrix} \theta & 1 \\ 1 & \theta \end{pmatrix}.$$

Dependent lifted functions are quotiented out. Every other nonzero eigenvector is already in this image. Thus its gap above constants is at least $(1 - \rho)/2 = 25/54$. A possible base eigenvalue -1 contributes a lifted eigenvalue -1 and causes no problem for this upper bound on the nonconstant spectrum. Put $\delta_0 = 25/54$. With normalized inner products,

$$\mathbb{E}_s \text{ uniform, } t \sim P(s, \cdot) a_s(1 - a_t) = \mu - \langle a, Qa \rangle \geq \delta_0 \mu(1 - \mu). \quad (4.1)$$

Indeed, the spectral bound gives $\langle a, Qa \rangle \leq \mu^2 + (1 - \delta_0) \text{Var}(a)$. Since $0 \leq a_s \leq 1$, we have $\text{Var}(a) \leq \mu(1 - \mu)$, yielding (4.1).

Put $c_L = 1/L$. The probability of selecting two Alice clones in a cluster with Alice fraction $a = j/L$ is

$$p_L(a) = \frac{j(j-1)}{L(L-1)} = \frac{a(a - c_L)}{1 - c_L}.$$

It satisfies $\sqrt{p_L(a)} \geq (a - c_L)_+$. For $a = 0$ this is immediate; for $a \geq c_L$, square both nonnegative sides and use

$$p_L(a) - (a - c_L)^2 = \frac{c_L(a - c_L)(a + 1 - c_L)}{1 - c_L} \geq 0.$$

For nonnegative A, B, c , $(A - c)_+(B - c)_+ \geq AB - c(A + B)$: if one factor is truncated, the right side is nonpositive; otherwise expand the left side. Since the transition is doubly stochastic, $\mathbb{E}[a_s + 1 - a_t] = 1$. Hence (4.1) implies

$$\mathbb{E} \sqrt{p_L(a_s)p_L(1 - a_t)} \geq \delta_0 \mu(1 - \mu) - \frac{1}{L} \geq \frac{16}{225} \quad (L \geq 32).$$

The final value follows exactly from $(25/54)(33 \cdot 67/10000) - 1/32 = 16/225$. Cauchy–Schwarz gives $\mathbb{E}[p_L(a_s)p_L(1 - a_t)] \geq 256/50625$. Original transitions are between distinct state clusters because the base graph has no loops, so their selected pairs are independent.

Let Z count original transitions $u \mapsto \pi(u)$ whose two a_u -vertices are Alice and whose two $b_{\pi(u)}$ -vertices are Bob. Under the unconditioned cover and clone choices,

$$\mathbb{E}Z \geq \frac{256}{50625}v - 2 \geq \eta v, \quad \eta := 1/200. \quad (4.2)$$

The subtraction of two covers the exceptional root cluster. The excess of the coefficient over $1/200$ is exactly $23/405000$, so $v \geq 35218$ suffices for (4.2).

4.2 Exposure concentration and conditioning

We first work under the *unconditioned* law: the local permutations are independent and uniform, and the clone placements are independent across state clusters and independent of the permutations. Fix the communication partition, including the root's analysis color. For the clone placement $\mathcal{R}_s$ in I_s , write

$$A_s = \mathbf{1}_{\{a_{s,0}, a_{s,1} \text{ are Alice}\}}, \quad B_s = \mathbf{1}_{\{b_{s,0}, b_{s,1} \text{ are Bob}\}}.$$

Then the count from (4.2) is

$$Z = \sum_{s \in S} A_s B_{\pi(s)}.$$

The auxiliary input bits are not involved in Z .

Exposure order and conditional means. Choose deterministic orders of the state clusters, base vertices, and neighbors of each base vertex. First reveal the v clone placements $\mathcal{R}_s$, one whole placement at a time. Then, at each base vertex w , reveal the first $d - 1$ images of σ_w in the chosen neighbor order; its last image is forced. Let $\mathcal{F}_i$ be the information after step i . There are $J = v + n(d - 1) \leq 2v$ steps, and

$$Y_i = \mathbb{E}[Z \mid \mathcal{F}_i], \quad Y_0 = \mathbb{E}Z, \quad Y_J = Z$$

is the corresponding Doob martingale. For a fixed positive-probability history through step $i - 1$, let $h_i(q)$ be the conditional expectation of Z when the next revealed value is q . We show that $|h_i(q) - h_i(q')| \leq 2$ for all compatible q, q' .

A clone-placement step. Compare two possible placements in I_s , and couple all unrevealed placements and all local permutations identically. This is a coupling of the correct conditional laws by independence. For each resulting fixed permutation π , only the summands indexed by s and $\pi^{-1}(s)$ can change: these are the unique outgoing and incoming transitions of the state. Each summand lies in $\{0, 1\}$, so the two completed values of Z differ by at most two. Averaging over this coupling gives the required bound for the two conditional means. The exclusion of r from its cluster's clone choices changes the allowed placements there, but not their independence from the remaining randomness.

A permutation-image step: the swap bijection. All clone placements are now fixed. Suppose the next image to be revealed is $\sigma_w(u)$. Let R be the set of unused images at w , and choose distinct $z, z' \in R$. Conditional on the previous images and on $\sigma_w(u) = z$, the remaining assignment is a uniform bijection onto $R \setminus \{z\}$. Given such a completed permutation, let $u' = \sigma_w^{-1}(z')$. The neighbor u' is unexposed and different from u . Swap the images at u, u' , obtaining

$$\sigma'_w(u) = z', \quad \sigma'_w(u') = z, \quad \sigma'_w(a) = \sigma_w(a) \quad (a \notin \{u, u'\}).$$

No previously exposed image changes, because neither z nor z' was used in the history. The same swap is its inverse. It is therefore a bijection from the completions with next image z to those with next image z' , and sends the uniform law on the first set to the uniform law on the second. Couple every other local permutation identically.

The corresponding state permutations differ only at (u, w) and (u', w) . Since all A_s, B_s are fixed, at most these two summands in Z change, each by at most one. If Z' is the value in the paired completion, it follows that

$$|h_i(z) - h_i(z')| = |\mathbb{E}[Z - Z' \mid \text{the fixed history and next image } z]| \leq \mathbb{E}[|Z - Z'| \mid \text{the same data}] \leq 2.$$

If only one image is available, the revelation is deterministic and has conditional range zero.

Conditional ranges and the tail constant. At any step, let m_i, M_i be the minimum and maximum of $h_i(q)$ over its compatible next values. The preceding couplings give $M_i - m_i \leq 2$. Conditional on $\mathcal{F}_{i-1}$, the martingale difference $D_i = Y_i - Y_{i-1}$ has mean zero and lies in the interval

$$[m_i - Y_{i-1}, M_i - Y_{i-1}],$$

whose length is at most two. Thus conditional Hoeffding's lemma, with the deterministic bounds $c_i = 2$, gives

$$\mathbb{E}[e^{\lambda D_i} \mid \mathcal{F}_{i-1}] \leq e^{\lambda^2 c_i^2 / 8}, \quad \sum_{i=1}^J c_i^2 = 4(v + n(d-1)) \leq 8v.$$

Iterating conditional expectations bounds $\mathbb{E}e^{-\lambda(Z - \mathbb{E}Z)}$ by $e^{\lambda^2 v}$. Markov's inequality and the choice $\lambda = t/(2v)$ consequently yield, for every $t > 0$,

$$\Pr[Z - \mathbb{E}Z \leq -t] \leq \exp\left(-\frac{t^2}{4v}\right).$$

Finally, (4.2) gives $\mathbb{E}Z \geq \eta v$. The event $Z < \eta v/2$ is therefore contained in $\{Z - \mathbb{E}Z \leq -\eta v/2\}$. Taking $t = \eta v/2$ gives

$$\Pr[Z < \eta v/2] \leq \exp\left(-\frac{2(\eta v/2)^2}{8v}\right) = \exp(-\tau v), \quad \tau := 1/640000. \quad (4.3)$$

The bounds are on conditional ranges, not merely on absolute martingale increments. This accounts for the constant in (4.3). Exposing a local permutation as a single block would lose a factor d and would not justify the conditioning argument at this degree.

The swap coupling was used only before conditioning on E ; it need not preserve the single-cycle event. We now use $\Pr[A \mid E] \leq \Pr[A]/\Pr[E]$, rather than assert the same coupling under the conditioned law. By (3.6),

$$\Pr[Z < \eta v/2 \mid E] \leq v \exp\left[-\left(\tau - \frac{27}{50d^2}\right)v\right]. \quad (4.4)$$

For every $d \geq 734$,

$$\frac{27}{50d^2} < \frac{1}{960000} = \frac{2\tau}{3}. \quad (4.5)$$

Thus (4.4) is at most $ve^{-\tau v/3}$. For $v \geq 2^{32}$, $\log v \leq v/3840000 = \tau v/6$; this follows at the endpoint from $2^{32} > 32 \cdot 3840000$, and from the decrease of $\log v/v$ thereafter. Consequently

$$\Pr[Z < \eta v/2 \mid E] \leq e^{-v/3840000}. \quad (4.6)$$

No independence after conditioning on E is asserted. Removing the single root stage loses at most one term. For $v \geq 800$, (4.6) leaves at least $\eta v/4 = v/800$ honoured active stages, except with the displayed failure probability.

4.3 Coefficient transfer and shared gates

Shared-gate decomposition

We give a first-principles proof of the balanced-product argument and its coefficient charge. These are the structural and rectangular decomposition tools used in [2, Theorem 2 and Lemma 9]; they are not claimed as new lower-bound methods. Write $A \preceq B$ when every coefficient of A is at most the corresponding coefficient of B .

Lemma 4.1 (Balanced products with shared gates). *Let $X_1, \dots, X_g$ be disjoint nonempty finite variable groups, where $g \geq 4$. Suppose a monotone arithmetic circuit with s binary arithmetic gates computes a nonzero full-group set-multilinear polynomial p . Then*

$$p = \sum_{j=1}^t U_j V_j, \quad t \leq s,$$

where U_j, V_j have nonnegative coefficients and are set-multilinear on complementary group sets $S_j, [g] \setminus S_j$, with $g/3 \leq |S_j| \leq 2g/3$. In particular, $U_j V_j \preceq p$. Constants may be arbitrary nonnegative reals, and fan-out is unrestricted. No condition on the intermediate polynomials is assumed.

Proof. *Productive gates have fixed group sets.* First propagate zero values and discard branches that cannot contribute to the output. A multiplication with a zero child is zero; a zero summand can be discarded. This introduces no arithmetic gates. Every remaining gate has a nonzero polynomial and a path to the output whose sibling at each multiplication is nonzero. Call such a gate productive, and write P_u for its polynomial.

Fix one such path from u to the output. At each multiplication on the path, choose a monomial of its sibling with positive coefficient. Addition preserves coefficientwise domination, and multiplication by a nonnegative polynomial preserves it as well. Induction along the path therefore gives

$$\lambda m P_u \preceq p$$

for a monomial m and a real $\lambda > 0$. This remains true if a sibling also depends on u : its chosen monomial is a monomial of its actual polynomial, and the same coefficientwise inequality applies.

For every monomial a of P_u , the monomial ma must thus use exactly one variable from every group, with exponent one. Consequently m uses a fixed subset of groups once, and every monomial of P_u uses exactly its complementary subset once. Denote the latter subset by $S(u)$. At an addition gate the nonzero children have the same group set as the parent. At a multiplication gate the two group sets are disjoint and their union is the parent's: multiply any nonzero monomial of each child to obtain these assertions.

A balanced arithmetic gate exists. Start at the output, whose group set has size g . While the current size exceeds $2g/3$, at an addition choose a nonzero child; at a multiplication choose a child whose group set has size at least $g/3$. Such a multiplication child exists because the two disjoint sizes sum to more than $2g/3$. Continue if the chosen child still has size greater than $2g/3$. Acyclicity and the fact that an input uses at most one group force this descent to encounter a gate u with $g/3 \leq |S(u)| \leq 2g/3$. Since $g \geq 4$, this gate uses at least two groups and is therefore an arithmetic gate.

Substitute at every use of that gate. Write $U = P_u$. Replace the value computed at node u by one fresh indeterminate z , at all outgoing uses simultaneously. Evaluate the resulting DAG formally to obtain

$$Q(x, z) = \sum_{k \geq 0} A_k(x) z^k, \quad A_k \in \mathbb{R}_{\geq 0}[X].$$

Substituting $z = U(x)$ restores the circuit, so $Q(x, U) = p$. The polynomial depends on z : the productive path from u propagates a monomial containing z , since all multiplication siblings remain nonzero polynomials after the replacement. Their substitution at $z = U$ gives their original nonzero values.

If some $A_k \neq 0$ for $k \geq 2$, choose monomials a of A_k and b of U with positive coefficients. Then ab^k occurs in $Q(x, U) = p$ with positive coefficient. It repeats every group of the nonempty set

$S(u)$, contradicting full-group set-multilinearity. Hence $A_k = 0$ for $k \geq 2$, and $A_1 \neq 0$. We have the exact identity

$$p = UA_1 + A_0, \quad A_0 = Q(x, 0).$$

For any monomial a of A_1 and any fixed monomial b of U , the product ab occurs in p . Therefore A_1 is set-multilinear on exactly $[g] \setminus S(u)$. Nonnegativity gives $UA_1 \preceq p$.

Remove nodes permanently and telescope. Apply the preceding argument to the current nonzero residual, starting with $R_0 = p$. At iteration j , take U_j to be the selected node's value in that residual circuit, and take V_j to be the coefficient of z after global replacement. Permanently set the selected node to zero. The resulting circuit computes R_j , with

$$R_{j-1} = U_j V_j + R_j, \quad 0 \preceq R_j \preceq R_{j-1} \preceq p.$$

Every nonzero residual is again full-group set-multilinear, so the argument can be repeated. Each iteration selects a different original arithmetic node: a node permanently set to zero cannot be productive again. Pruning does not add nodes. Thus there are at most s iterations, and the final residual is zero; otherwise the balanced descent would supply another eligible node. Telescoping proves the stated decomposition and coefficient domination.

The count is of removed DAG nodes. No bound on the number of parse trees, paths, or uses of a node is needed. All their multiplicities are already included in the coefficient $V_j = A_1$. Neither V_j nor the displayed decomposition is required to have a separate small-circuit representation. $\square$

Proposition 4.2 (Coefficient charge for a signed measure). *Let $\Omega_i = X_i$ index the choices in group i , and put $\Omega = \prod_{i=1}^g \Omega_i$. Let $\mu : \Omega \rightarrow \mathbb{R}$ be a signed weight function such that, for every balanced partition $S, [g] \setminus S$ and every rectangle $A \times B$ in that partition,*

$$\left| \sum_{\nu \in A \times B} \mu(\nu) \right| \leq \gamma.$$

For a full-group polynomial define $\mathcal{M}_\mu(p) = \sum_{\nu \in \Omega} \mu(\nu) [m_\nu]p$. If p satisfies Lemma 4.1 and its coefficients are at most C , then

$$|\mathcal{M}_\mu(p)| \leq sC\gamma.$$

No product-distribution assumption on μ is made.

Proof. Fix one product UV in the lemma, with complementary group sets S, T . Let u_α, v_β be its factors' coefficient vectors over Ω_S, Ω_T , including zero coefficients. If $UV = 0$, its charge is zero, so assume both factors are nonzero. Disjoint groups give

$$[m_{(\alpha, \beta)}]UV = u_\alpha v_\beta.$$

Because the domain is the full Cartesian product and $UV \preceq p$,

$$\left(\max_{\alpha} u_{\alpha} \right) \left(\max_{\beta} v_{\beta} \right) = \max_{\alpha, \beta} u_{\alpha} v_{\beta} \leq C.$$

This bounds the product of the two maxima; neither individual maximum is assumed bounded by C .

List the distinct positive levels of u as $0 = a_0 < a_1 < \dots < a_r$, and those of v as $0 = b_0 < b_1 < \dots < b_\ell$. Define

$$A_i = \{\alpha : u_{\alpha} \geq a_i\}, \quad B_j = \{\beta : v_{\beta} \geq b_j\}, \quad \lambda_{ij} = (a_i - a_{i-1})(b_j - b_{j-1}) > 0.$$

The identities

$$u_\alpha = \sum_{i=1}^r (a_i - a_{i-1}) \mathbf{1}_{\{u_\alpha \geq a_i\}}, \quad v_\beta = \sum_{j=1}^\ell (b_j - b_{j-1}) \mathbf{1}_{\{v_\beta \geq b_j\}}$$

give the following entrywise equality on the full Cartesian domain:

$$u_\alpha v_\beta = \sum_{i=1}^r \sum_{j=1}^\ell \lambda_{ij} \mathbf{1}_{A_i \times B_j}(\alpha, \beta).$$

Each $A_i \times B_j$ is a rectangle for the same balanced partition S, T . Telescoping the level increments gives exactly

$$\sum_{i,j} \lambda_{ij} = \left(\sum_i (a_i - a_{i-1}) \right) \left(\sum_j (b_j - b_{j-1}) \right) = a_r b_\ell = (\max u)(\max v) \leq C.$$

Thus it is the total scalar weight, regardless of how many rectangles occur, that controls the charge:

$$|\mathcal{M}_\mu(UV)| = \left| \sum_{i,j} \lambda_{ij} \sum_{\nu \in A_i \times B_j} \mu(\nu) \right| \leq \gamma \sum_{i,j} \lambda_{ij} \leq C\gamma.$$

This is the rectangular decomposition of [2, Lemma 9, pp. 39:10–39:11], followed by the signed-measure estimate in its Lemma 7. The displayed proof allows arbitrary nonnegative real coefficients and an arbitrary signed weight function μ . Sum this inequality over the at most s products in Lemma 4.1. $\square$

Discrepancy and coefficient transfer

Fix a balanced partition, condition on the layout, and fix all unhonoured bits. By locality, the pullback of any communication rectangle is an inner-product rectangle in the remaining bits. For the 2^z -by- 2^z inner-product sign matrix A , the identity $AA^\top = 2^z I$ gives norm $2^{z/2}$. For rectangle indicator vectors, Cauchy–Schwarz after normalization by 2^{2z} bounds the signed mass by $2^{-z/2}$. Using (4.6) and $z \geq v/800$, the original distribution has

$$\gamma_0 \leq e^{-v/3840000} + 2^{-v/1600}. \quad (4.7)$$

This is the same distribution for every balanced partition. No union bound over partitions or simultaneously good layout is needed.

Every layout has $k = v - 1$ bits of inner product. Its two answer masses are $(1 - 2^{-k})/2$ and $(1 + 2^{-k})/2$. Balance them exactly, changing total variation by 2^{-k-1} . Explicitly, if $p_i = \Pr_\Delta[g_H = i]$, set $\Delta_b(\nu) = \Delta(\nu)/(2p_{g_H(\nu)})$. Then both answers have mass $1/2$. Every rectangle's signed expectation changes by at most 2^{-k} . For $v \geq 2^{32}$, the centered maximum discrepancy is therefore

$$\gamma \leq 2^{-cv}, \quad c := 1/7680000. \quad (4.8)$$

Indeed the left side is at most $3 \cdot 2^{-v/3840000}$, and the factor three is absorbed by half the exponent.

Define $M(m_\nu) = \Delta_b(\nu)(2g_H(\nu) - 1)$. Then

$$M(F_H) = 0, \quad M(P_H) = 1/2, \quad |M(F_H \pm \epsilon P_H)| = \epsilon/2.$$

To make the coefficient bound in Proposition 4.2 explicit here, put $\mu(\nu) = \Delta_b(\nu)(2g_H(\nu) - 1)$ and $p = F_H \pm \epsilon P_H$. For each balanced product $U_j V_j \preceq p$ supplied by Lemma 4.1, every coefficient satisfies

$$0 \leq u_\alpha v_\beta \leq [m_{(\alpha, \beta)}]p \leq 1 + \epsilon \leq \frac{3}{2}.$$

The level-set rectangles in Proposition 4.2 therefore have total weight

$$\sum_{i,k} \lambda_{ik} = (\max_\alpha u_\alpha)(\max_\beta v_\beta) = \max_{\alpha, \beta} u_\alpha v_\beta \leq \frac{3}{2}.$$

Each rectangle has signed mass at most γ in absolute value by (4.8), so $|M(U_j V_j)| \leq (3/2)\gamma$. The bound holds even if one factor has very large coefficients: only the product of the two maxima is used. Summing over $t \leq s$ products gives

$$\frac{\epsilon}{2} = |M(p)| \leq \sum_{j=1}^t |M(U_j V_j)| \leq \frac{3}{2} t \gamma \leq \frac{3}{2} s \gamma,$$

and hence

$$s \geq \frac{\epsilon}{3\gamma}. \tag{4.9}$$

Since $N < Kv$, $\epsilon \geq 2^{-aN}$ with $a = c/(2K)$ implies $\epsilon \geq 2^{-cv/2}$. Thus

$$s \geq 2^{cv/2}/3 \geq 2^{cv/3} \geq 2^{bN}$$

with $b = c/(3K)$; the middle inequality is already valid under the stated cutoff. This proves Theorem 2.1.

5 Explicit families and actual variable counts

5.1 An explicit LPS family

Fix the prime $p = 733$. For every field prime

$$q \equiv 1 \pmod{2932}, \quad q > 1466,$$

the standard LPS graph [9] B_q is the nonbipartite Cayley graph on $\text{PSL}_2(\mathbb{F}_q)$ of degree $d = p + 1 = 734$. It has

$$n = \frac{q(q^2 - 1)}{2}, \quad v = 367q(q^2 - 1).$$

Indeed $p, q \equiv 1 \pmod{4}$, and quadratic reciprocity plus $q \equiv 1 \pmod{p}$ gives $(p/q) = 1$. The LPS spectral theorem gives

$$\max_{i \geq 2} |\theta_i| \leq \frac{2\sqrt{733}}{734} < \frac{2}{27}.$$

Connectivity and nonbipartiteness hold on this PSL component. The conservative bound $q > 2p$ ensures the standard generators remain nonidentity and distinct modulo q , so the graph is simple. No girth theorem is used.

One can construct the fixed generator list from integer quaternions (a_0, a_1, a_2, a_3) of norm 733, where a_0 is positive and odd and the other coordinates are even. With $i^2 = -1$ modulo q , send it to the projective matrix

$$\begin{pmatrix} a_0 + ia_1 & a_2 + ia_3 \\ -a_2 + ia_3 & a_0 - ia_1 \end{pmatrix}.$$

Its determinant is 733, a square modulo q ; normalizing to determinant one gives a PSL element independent of the sign of the square root. This is the conventional LPS construction. For the fixed p , exhaustive enumeration of the generator list and of determinant-one matrices modulo $\{\pm I\}$ takes time polynomial in the graph size.

For clarity, simplicity can also be checked directly in this single-prime construction. Normalize every matrix by the same square root of 733. Equality in PSL then means equality or negated equality of the two normalized matrices. The quaternion matrix map is invertible over $\mathbb{F}_q$, so the coordinate vectors are congruent up to the same sign. Since every coordinate has absolute value at most $\sqrt{733}$ and $q > 2 \cdot 733$, the congruence is an integer equality. The positive a_0 convention eliminates the minus case. An identity generator would require $a_1 = a_2 = a_3 = 0$, contradicting $a_0^2 = 733$. Quaternion conjugation gives inverse closure of the generator list.

Take $L = 32$. The exact host degree and variable factor are

$$D_H = 17263679, \quad K = LD_H = 552437728.$$

With $c = 1/7680000$, the constants in the theorem can be chosen as the positive rationals

$$a = \frac{1}{8485443502080000}, \quad b = \frac{1}{12728165253120000}.$$

The cutoff $v \geq 2^{32}$ is already implied by $q > 1466$. All these constants are deliberately conservative.

5.2 Every sufficiently large variable count

The core graph theorem gives an explicit infinite subsequence. We give a padding argument for all sufficiently large actual variable counts. It uses only the prime number theorem in the fixed progression $1 \pmod{2932}$, in addition to the core graph result.

Set $m = D_H$, $A = 367L$. The core number of groups is

$$g_q = Aq(q^2 - 1) - 1.$$

For any sufficiently large integer N , let $g = \lfloor N/m \rfloor$ and $x = (g/A)^{1/3}$. The prime number theorem in this fixed progression gives an admissible prime $q \in (x/2, x]$ once x is large enough; for a stronger quantitative statement see [12]. Then $g_q \leq g$. Since $q \geq 2$ and $q^2 - 1 \geq 3q^2/4$,

$$g_q \geq \frac{3g}{32} - 1 \geq \frac{g}{16} \quad (g \geq 32).$$

The core variable count $N_0 = mg_q$ is therefore at least $N/32$, using $mg \geq N/2$ for $N \geq 2m$.

Append $g - g_q$ fresh groups of size m , and, if $N - mg > 0$, one final fresh group of size $N - mg$. Let G be the product of all appended group sums, with $G = 1$ if none are needed. Define

$$F_N = F_{H_q}G, \quad P_N = P_{H_q}G.$$

These are explicit polynomials on exactly N variables. Fixing one variable in each appended group to one and the others to zero restricts $F_N \pm \epsilon P_N$ to the core target without increasing monotone circuit size.

For $a' = a/32$ and $b' = b/32$, the conditions

$$2^{-a'N} \leq \epsilon \leq 1/2$$

imply $\epsilon \geq 2^{-aN_0}$, since $N_0 \geq N/32$. The core theorem therefore gives size at least $2^{bN_0} \geq 2^{b'N}$. The finitely many smaller indices may be filled arbitrarily. Primality testing or direct enumeration in this fixed progression is polynomial in N ; the determinant construction stays uniform.

This does not assert a universal discrepancy estimate for every balanced partition of the padded variables. The lower bound is transferred by an arithmetic restriction after the shared-gate theorem has already been proved on the core.

This completes the proof of Theorem 1.1, with its constants taken to be a', b' .

6 Full support and sharp approximation

6.1 Two coefficient values

The polynomial $R_H = 2F_H - P_H$ has every full-group monomial, each with coefficient one or two. It is in uniform VP over the integers. A monotone circuit of size s for R_H , followed by one multiplication by $1/2$, computes $F_H - \frac{1}{2}P_H$. Thus $s + 1 \geq 2^{bN}$, implying $s = 2^{\Omega(N)}$. The padded family has the same conclusion.

More generally, let $Q = F_H \pm \epsilon P_H$. Suppose $\tilde{Q}$ is a nonnegative set-multilinear polynomial on exactly the full set of tail groups and

$$|[m_\nu]\tilde{Q} - [m_\nu]Q| \leq \epsilon/4 \quad \text{for every neighbor map } \nu.$$

The centered coefficient functional has total absolute weight one, so $|M(\tilde{Q})| \geq \epsilon/4$. Its coefficients are at most $1 + \epsilon + \epsilon/4 < 2$. The same balanced-product argument gives

$$\text{monSIZE}(\tilde{Q}) \geq \frac{\epsilon}{8\gamma}.$$

This is a strongly exponential coefficient-approximation lower bound in the same perturbation range, after decreasing the exponent constant if needed. The requirement that the output be set-multilinear on exactly these groups excludes uncontrolled extra monomials; it is not an approximation claim for arbitrary nonhomogeneous output.

Conversely, the number of full-group monomials is $D_H^{Lv-1} = 2^{O(N)}$. Listing them gives a monotone upper bound $2^{O(N)}$ for the exact targets. The size scale is therefore $2^{\Theta(N)}$ with fixed, different constants.

6.2 Exact coefficient-approximation tradeoff

Use the same host, full product F_H , arborescence polynomial P_H , and exactly centered distribution Δ_b . Let γ bound its rectangle discrepancy for every balanced partition. The Eulerian theorem gives $\gamma \leq 2^{-cv}$. The signed coefficient functional

$$M(Q) = \sum_{\nu} \Delta_b(\nu)(2g_H(\nu) - 1)[m_\nu]Q$$

assigns total positive weight $1/2$ and total negative weight $-1/2$.

Fix $0 < \ell < u$. Consider either assignment of the two coefficient levels to the two answers:

$$R_{\ell,u} = \ell P_H + u(F_H - P_H) \quad \text{or} \quad R_{\ell,u} = u P_H + \ell(F_H - P_H).$$

Let $1 \leq \kappa < \sqrt{u/\ell}$. Suppose a monotone circuit computes a full-group set-multilinear polynomial $\tilde{R}$ whose coefficients satisfy

$$\frac{[m_\nu]R_{\ell,u}}{\kappa} \leq [m_\nu]\tilde{R} \leq \kappa[m_\nu]R_{\ell,u} \quad \text{for every } \nu. \quad (6.1)$$

Then its size satisfies the explicit bound

$$\boxed{s \geq \frac{u - \kappa^2 \ell}{2\kappa^2 u \gamma}.} \quad (6.2)$$

Proof. First take the assignment with level ℓ on $g_H = 1$ and level u on $g_H = 0$. Exact centering and (6.1) give

$$\begin{aligned} M(\tilde{R}) &\leq \frac{\kappa \ell}{2} - \frac{u}{2\kappa} \\ &= -\frac{u - \kappa^2 \ell}{2\kappa} < 0. \end{aligned}$$

The other assignment reverses the sign and gives the same lower bound on $|M(\tilde{R})|$. Every output coefficient is at most κu . The shared-gate balanced-product decomposition therefore writes the output as at most s terms, each having rectangle-mixture mass at most κu . Each term's functional magnitude is at most $\kappa u \gamma$. Comparing the two bounds yields (6.2). No precision restriction on the constants is used. $\square$

For fixed ℓ, u, κ with $\kappa < \sqrt{u/\ell}$, the factor preceding $1/\gamma$ is a positive constant. Consequently (6.2) is $2^{\Omega(N)}$ in actual variables. If κ depends on N and approaches the threshold, only the explicit tradeoff (6.2), rather than a uniform exponent, is being asserted.

6.3 Matching easy approximation at the threshold

At $\kappa_0 = \sqrt{u/\ell}$, the easy polynomial

$$\sqrt{\ell u} F_H \quad (6.3)$$

satisfies (6.1) for both coefficient levels. For the lower level, $\sqrt{\ell u} = \kappa_0 \ell$; for the higher level, $\sqrt{\ell u} = u/\kappa_0$. Thus (6.3) has an $O(N)$ -size monotone circuit and reaches the threshold exactly in the real-constant model.

In particular, for

$$R_H = 2F_H - P_H,$$

the symmetric multiplicative coefficient-approximation threshold is $\sqrt{2}$: for every fixed $1 \leq \kappa < \sqrt{2}$,

$$s \geq \frac{2 - \kappa^2}{4\kappa^2 \gamma} = 2^{\Omega(N)}, \quad (6.4)$$

whereas $\sqrt{2}F_H$ is an $O(N)$ -size $\sqrt{2}$ -approximation. The lower bound is for unrestricted monotone circuits with shared gates, not just sums of products.

6.4 Uniform pointwise approximation needs no group-structure assumption

The same sharp threshold holds if an arbitrary monotone circuit is required to approximate $R_{\ell,u}$ pointwise even on the bounded cube:

$$\frac{R_{\ell,u}(x)}{\kappa} \leq \tilde{R}(x) \leq \kappa R_{\ell,u}(x) \quad \text{for every } x \in [0, 1]^N. \quad (6.5)$$

No degree or set-multilinearity assumption on the circuit's output or intermediate gates is needed. The following standard extraction argument [7, 8] preserves size even when gates are shared.

Lemma 6.1 (Least homogeneous component). *If a nonzero polynomial Q is computed by a size- s monotone circuit, its nonzero homogeneous component of least total degree has a monotone circuit of size at most s .*

Proof. Prune zero and unused gates. Assign each remaining gate its least monomial degree. In addition this is the minimum of the children's degrees; at multiplication it is their sum, because no cancellation is possible. Build one corresponding gate for each original gate. At an addition with unequal least degrees retain only the child of smaller degree; at equal degrees retain their addition. At multiplication retain the product of the two least components. Inputs remain variables or nonnegative constants. Induction computes each gate's least homogeneous component. Each original gate contributes at most one arithmetic gate, and all of its uses share that same constructed gate. Thus size does not increase. $\square$

Let g be the number of tail groups, so the target $R_{\ell,u}$ is homogeneous of degree g and positive at $\mathbf{1}$. Let d be the least degree of $\tilde{R}$. Substitute $x = t\mathbf{1}$ in (6.5), divide by t^g , and let t decrease to zero. If $d < g$, the upper bound fails; if $d > g$, the lower bound fails. Hence $d = g$.

Let Q_g be that least homogeneous component. It has monotone size at most the size of the original approximator. For every fixed $x \in [0, 1]^N$, the points tx remain in the cube for $0 < t \leq 1$. Dividing (6.5) at tx by t^g and taking the limit gives

$$R_{\ell,u}(x)/\kappa \leq Q_g(x) \leq \kappa R_{\ell,u}(x).$$

On a face where every variable of one tail group is zero, the target vanishes. Nonnegativity therefore forces every monomial of Q_g to meet every tail group. Its degree is exactly g , so it uses each group exactly once. Thus Q_g is full-group set-multilinear.

Evaluate at the one-hot point for each neighbor map: set its selected variable in every group to one and all others to zero. This isolates the corresponding coefficient of Q_g , proving (6.1) for Q_g . Applying (6.2) to its no-larger circuit proves the same bound for the original arbitrary monotone approximator.

Conversely, a full-group polynomial satisfying (6.1) satisfies the pointwise inequalities on the whole nonnegative orthant by summing termwise. In particular (6.3) attains the threshold on the cube and on the orthant. The optimal monotone size for coefficient approximation equals that for uniform cube approximation, with no size loss in either direction. The latter assertion compares the optimal circuit sizes; it does not say that every cube approximator itself is set-multilinear.

For example $xy + (xy)^2$ is a factor-two approximation to xy on $[0, 1]^2$ and is not multilinear. The least-component operation extracts xy . This explains why forcing the original output to be multilinear would have been too strong, while the size lower bound still transfers.

The same conclusion holds if approximation is required on the open positive cube, since polynomial continuity supplies its boundary. It also holds on the entire nonnegative orthant. The argument uses the continuum of scales tending to zero; it does not give an approximation theorem on Boolean inputs alone.

6.5 Padding and scope

The all-size padding in Section 5.2 preserves this corollary. Choosing one variable in each added group to be one and the others zero isolates one padded extension of each core monomial. Coefficient

approximation restricts to coefficient approximation on the core; pointwise approximation restricts directly. A constant fraction of the actual variables remains.

The two-level target has a polynomial-size general arithmetic circuit because P_H is a rooted determinant. At levels one and two, no growing coefficient precision is needed even in the exact target. This is a sensitive monotone arithmetic separation and its sharp approximation corollary.

The Boolean support function of R_H is the same easy support function as F_H . This result does not imply a monotone Boolean circuit lower bound through a support conversion. It also does not supply Hrubeš's additional hypotheses for general arithmetic lower bounds.

7 Exact counting on the Boolean cube

Exact counting requires equality with the target on every Boolean assignment, in the sense of [8]. The full-group normalization below is related to the consistent-to-decomposable normalization of sum-product networks [11, 10]. We include its proof to specify the hypotheses, arbitrary-real-constant model, and cost in shared arithmetic gates.

7.1 Counting normalization statement

Partition the variables into $g \geq 1$ nonempty groups $X_1, \dots, X_g$. Let $p \neq 0$ be a polynomial with nonnegative coefficients whose every monomial uses exactly one variable from each group, with exponent one. Suppose a size- s monotone arithmetic circuit computes some polynomial Q satisfying

$$Q(x) = p(x) \quad (x \in \{0, 1\}^N). \quad (7.1)$$

The circuit has binary $+$ and $\times$ gates, arbitrary nonnegative real constants, and unrestricted reuse; Q need not be multilinear or homogeneous.

Theorem 7.1 (Full-group counting normalization). *There is a monotone circuit computing p as a formal polynomial, of size at most*

$$(2g + 1)s + g. \quad (7.2)$$

The transformation preserves shared subcircuits and takes polynomial time in the original circuit and variable count. It does not depend polynomially on the potentially huge formal degree of Q .

7.2 Profiles of productive gates

For a polynomial U , write $\text{lin}(U)$ for the result of replacing each positive exponent by one and combining like terms. A multilinear polynomial is uniquely determined by its values on the Boolean cube; this follows inductively by writing it as $A + x_N B$ and evaluating at $x_N = 0, 1$. Thus (7.1) implies

$$\text{lin}(Q) = p. \quad (7.3)$$

Nonnegativity then implies that every monomial of Q has exactly one distinct variable from each group, although that variable may have a higher exponent.

Prune zero and unused gates. For each productive gate v , let $S_i(v) \subseteq X_i$ be the set of variables from group i appearing in any monomial at v . Call i flexible at v if $|S_i(v)| \geq 2$, and fixed at v if $|S_i(v)| = 1$. Empty profiles are allowed.

Every monomial at v is a partial choice: it cannot contain two distinct variables from a group. Indeed choose a nonzero output parse context containing an occurrence of v , and fix monomials in all its other factors and occurrences. Every monomial at the designated occurrence, times this fixed context, occurs at the output with a positive coefficient. Two different variables of a group could not disappear.

Moreover, every flexible group at v occurs in every monomial at v . For the same fixed context, if it contained a variable from such a group, some monomial at v with a different variable would create an invalid output support. Hence the context contains no variable from that group. The full-group output condition then forces every monomial at v to supply it.

At a multiplication gate $v = ut$, if both children have a nonempty profile in a group, both profiles must be the same singleton. Otherwise a cross-product of monomials would contain two different variables from that group. Consequently the flexible groups at v are the disjoint union of the flexible groups at its two children.

At an addition gate $v = u + t$, profiles are their unions. If a group is flexible at v but not flexible at a nonzero child, that child has a singleton profile there, and its variable occurs in every child monomial. An empty profile or an omitting monomial would contradict the mandatory presence of every flexible group at v .

7.3 Constructing the normalized circuit

Define J_v by setting all variables of fixed groups at v to one, then linearizing the remaining polynomial. The preceding facts show that J_v is set-multilinear on exactly the flexible groups at v .

For an input variable, $J_v = 1$. For a constant input, J_v is that constant.

At a multiplication gate the flexible group sets are disjoint, and every other present group is fixed at both relevant children and parent. Hence

$$J_v = J_u J_t. \tag{7.4}$$

This uses one multiplication gate.

At an addition gate, let M_u be the product of the unique child variables from groups that are flexible at v and fixed at u ; define M_t analogously. Each is a product of at most g distinct variables. The mandatory-presence property shows that restoring one copy of each such variable exactly reverses its removal under linearization. Therefore

$$J_v = M_u J_u + M_t J_t. \tag{7.5}$$

Each restored product can be formed by multiplying its variables into the already computed child output. This costs at most $2g$ multiplications and one addition per original addition gate.

Construct one normalized output for each original gate and reuse it at every parent. Finally, at the output, restore one copy of the unique variable from each globally fixed group. All these groups are mandatory by (7.3). The result is exactly p , at cost at most g more multiplications. The total size is bounded by (7.2).

Profiles are computed from the input variables by set union at both addition and multiplication. The latter union rule uses the fact that all productive polynomials are nonzero and no cancellation occurs. All other operations of the construction are explicit. This proves the theorem. $\square$

7.4 Consequence for the sensitive VP family

Let $C_{01}(T)$ denote monotone circuit size required only to agree with T on the entire Boolean cube. Apply the normalization theorem to the full-group target $T = F_H \pm \epsilon P_H$. The coefficient discrepancy bound in the Eulerian proof gives

$$C_{01}(T) \geq \frac{\epsilon/(3\gamma) - g}{2g + 1}, \quad g = |V(H)| - 1, \quad \gamma \leq 2^{-cv}. \quad (7.6)$$

In particular, for the same fixed host parameters and $2^{-aN} \leq \epsilon \leq 1/2$,

$$C_{01}(T) = 2^{\Omega(N)} \quad (7.7)$$

for sufficiently large N . The polynomial normalization factor is absorbed into the exponent. This statement does not claim the identical numerical exponent at the original finite cutoff for arbitrary fixed clone counts.

At $\epsilon = 1/2$, the two-coefficient target $2F_H - P_H$ has the same strongly exponential exact counting lower bound. Its values on Boolean inputs count weighted full-group monomials. This is stronger than requiring the competing circuit to compute the formal polynomial, while still being an arithmetic counting model, not a Boolean decision model.

The normalizer uses (7.3), which follows from exact equality on all Boolean assignments. Approximation, or equality only on one-hot assignments, does not supply (7.3). In particular, an approximator may contain monomials using two distinct variables from one group. No Boolean approximation theorem is inferred here.

8 Scope and further questions

The host polynomial has full support within its full-group monomial basis. Replacing arithmetic addition and multiplication by Boolean OR and AND therefore produces the same easy support predicate as for F_H . The lower bound concerns arithmetic coefficients and does not give a Boolean circuit lower bound by a support conversion.

The distinction between complete input domains and promises is also substantive. On the promise of exactly one selected variable in every outgoing group, the targets have polynomial-size monotone arithmetic circuits. To see this, represent the complement of $x_{z,w}$ on the promise by the sum of the other variables in its group. Prefix and suffix sums produce all these complements in $O(N)$ gates. A Boolean reachability circuit can now be simulated with two nonnegative arithmetic outputs $a, \bar{a}$ for each Boolean gate. The identities

$$\begin{aligned} \text{AND: } & ab, & \bar{a} + \bar{a}\bar{b}, \\ \text{OR: } & a + \bar{a}\bar{b}, & \bar{a}\bar{b} \end{aligned}$$

compute a gate and its complement whenever the incoming pairs are complementary bits; NOT merely exchanges the outputs. Reachability for every vertex uses polynomially many Boolean gates. If $A, \bar{A}$ are the resulting arborescence-indicator pair, then $1 + \bar{A}$, $1 + \epsilon A$, and $(1 - \epsilon) + \epsilon \bar{A}$ compute respectively $2F_H - P_H$, $F_H + \epsilon P_H$, and $F_H - \epsilon P_H$ on the promise, since $F_H = 1$ there. These circuits need not compute the formal polynomials or their values on the entire Boolean cube. Multiplying by F_H can enforce vanishing on all group-zero faces without changing the promised values.

Division is another distinct model. Subtraction-free rational algorithms for arborescence polynomials are known [5]. Our lower bound is for division-free circuits and does not preclude such

algorithms. Similarly, the perturbation interval in Theorem 2.1 does not supply the additional hypotheses in Hrubeš’s criteria for general arithmetic lower bounds [6]. In particular, we do not obtain a separation of VP from VNP or of P from NP.

Several questions remain. Can the required base degree and clone count be substantially reduced? Can the sensitive size bound be extended to smaller perturbations? Does a corresponding approximation bound hold when the domain is only the Boolean cube? Finally, can strongly exponential sensitive hardness be achieved for a family with polynomial-size general arithmetic formulas? The determinant upper bound here supplies polynomial-size circuits, not that stronger formula upper bound.

Acknowledgment

The author acknowledges OpenAI GPT-6 for assistance with research exploration, proof development, verification code, and manuscript preparation. The finite computations are supplementary checks; the asymptotic claims rest on the written arguments.

A Direct BEST counting and sampling

The following standard last-exit-tree proof [4] is included to fix the counting convention. Replace every base edge by its two directed arcs. Fix a base start vertex o . Choose a directed spanning tree toward o . At each $u \neq o$, order its outgoing arcs with the tree arc last, and order the other $d - 1$ arcs arbitrarily. At o , choose any order of its d outgoing arcs. Starting at o , always take the next unused outgoing arc in that vertex’s order.

The walk cannot stop away from o , by balance of indegree and outdegree. If it stops at o with an unused outgoing arc somewhere, follow unused last-exit tree arcs toward o . The first fully exhausted vertex on that tree path would have an unused incoming arc, contradicting balance for the closed walk. Thus all arcs are used, and the walk is an Eulerian tour.

Conversely, the last outgoing arcs of nonroot vertices in an Eulerian tour form a tree toward o . A directed cycle of these last exits would force their exit times to increase strictly around the cycle, which is impossible. The outgoing orders and last-exit tree are uniquely recovered from the tour.

There are therefore

$$t_o(B) d! ((d - 1)!)^{n-1}$$

linear Eulerian tours starting at o . Each cyclic transition configuration is represented by exactly d such tours, one for each possible first outgoing arc at o . Dividing by d gives $t_o(B)((d - 1)!)^n$, proving (3.2).

This also gives an efficient uniform sampler: choose a uniform spanning tree of B , orient it toward o , choose all the allowed outgoing orders uniformly, run the tour, and forget its initial position. Every single-cycle transition configuration has the same number d of preimages. A uniform spanning tree can, for example, be sampled in expected polynomial time by successive edge inclusion decisions using exact matrix-tree counts and deletion/contraction. Those counts have polynomial bit length and require polynomially many determinant computations. Sampling the exact rational branch probabilities takes expected polynomially many random bits. No rejection sampling with exponentially small acceptance is needed.

B A smaller physical host

Let S be the oriented base edges and define a simple undirected graph T_B on S by

$$\{(u, w), (x, z)\} \in E(T_B) \iff w = x \text{ or } z = u.$$

There are no loops because B is simple. For $s = (u, w)$, there are d successor states (w, z) and d predecessor states (z, u) . Their intersection is exactly $\{(w, u)\}$, so T_B has degree $2d - 1$. It is connected, for example because any Euler tour of the bidirected connected base traces a spanning cycle in T_B .

The host

$$H^\circ = T_B[K_L]$$

has Lv physical vertices and constant degree

$$D^\circ = 2dL - 1.$$

Every intercluster arc in the direct gadget goes from an oriented edge state to one of its successor states, and every other arc stays in a state cluster. Consequently every possible gadget assignment lies in H° .

The entire proof of Sections 3 and 4 therefore applies verbatim to this smaller host: its probability space, fractional partition calculation, exposure concentration, parity identity, total-domain rectangle argument, and shared-gate coefficient transfer do not use any additional host edges. In particular, with actual variable count

$$N^\circ = (Lv - 1)D^\circ,$$

one may take the same $c = 1/7680000$ and

$$a^\circ = \frac{c}{2LD^\circ}, \quad b^\circ = \frac{c}{3LD^\circ}.$$

For $v \geq 2^{32}$ and $2^{-a^\circ N^\circ} \leq \epsilon \leq 1/2$,

$$\text{monSIZE}(F_{H^\circ} \pm \epsilon P_{H^\circ}) \geq 2^{b^\circ N^\circ}.$$

This is a direct adaptation of the proved discrepancy argument, not a claim that deleting edges from an arbitrary hard polynomial preserves hardness. The same hard distribution is supported in the smaller total Cartesian domain, and that fact is what allows the transfer to be repeated.

For $d = 734, L = 32$, the degree drops from 17263679 to 46975, while the number of physical vertices is unchanged. The main statement keeps the simpler notation $B[K_{dL}]$; this optional host gives better fixed constants. Neither change alters the asymptotic $2^{\Theta(N)}$ scale.

C General conditioning and fixed-gap expanders

C.1 The finite conditioning criterion

Let B be a connected simple d -regular graph with n vertices and normalized second eigenvalue at most $0 \leq \rho < 1$. Put $v = dn$, fix $L \geq 5$, and use the direct gadget on $H = B[K_{dL}]$. Write

$$\delta = \frac{1 - \rho}{2}, \quad q_v = \frac{Lv - 1}{3Lv}, \quad w_v = \left(\delta q_v (1 - q_v) - \frac{1}{L} \right)_+^2 - \frac{2}{v}, \quad \beta = \frac{1}{2(1 - \rho)d^2}.$$

For every real t satisfying $0 < t < w_v$, the exactly centered distribution of the Eulerian construction obeys

$$\boxed{\max_{\text{balanced } A} \text{disc}_{\Delta_b, A}(g_H) \leq v \exp \left[- \left(\frac{(w_v - t)^2}{4} - \beta \right) v \right] + 2^{-(tv-1)/2} + 2^{1-v}.} \quad (\text{C.1})$$

The maximum is over partitions of whole outgoing-choice groups with each side between one third and two thirds of the groups. The predicate and rectangles are on the full Cartesian product of all neighbor choices.

Proof. Fix any such partition and assign the root either analysis color. Its mean Alice density lies in $[q_v, 1 - q_v]$. The state-transition gap is at least δ , with $0 < \delta \leq 1/2$. The pair estimate in Section 4.1 therefore gives

$$\mathbb{E}Z \geq w_v v.$$

The subtraction of two is solely the expectation loss at the root cluster's incoming and outgoing transition. Clones remain independent across clusters before conditioning.

Expose the v clone choices, then the local permutation images individually. The sum of squared conditional range lengths is at most $8v$. Hence

$$\Pr[Z < tv] \leq \exp \left[- \frac{(w_v - t)^2}{4} v \right].$$

This use of the mean lower bound is valid because $t < w_v$. The BEST count and logarithmic trace estimate give

$$\Pr[E] \geq v^{-1} e^{-\beta v}.$$

Dividing the preceding tail by this probability proves the first term of (C.1). On a remaining layout there are at least $tv - 1$ honoured active stages, after removing the root stage. Fixing all other bits pulls every rectangle back to an inner-product rectangle, proving the second term. There are exactly $v - 1$ inner-product stages on every layout. Exact answer centering changes signed rectangle expectation by at most 2^{1-v} , giving the final term.

All these estimates hold for each fixed partition under the same distribution. No union bound over partitions, post-conditioning independence, or promised input domain has been introduced. $\square$

As v tends to infinity with fixed parameters,

$$w_v \longrightarrow w_\infty = \left(\frac{1 - \rho}{9} - \frac{1}{L} \right)_+^2.$$

Thus a sufficient asymptotic condition is

$$\boxed{\frac{1}{2(1 - \rho)d^2} < \frac{1}{4} \left(\frac{1 - \rho}{9} - \frac{1}{L} \right)_+^4.} \quad (\text{C.2})$$

Indeed choose a fixed t with $0 < t < w_\infty - 2\sqrt{\beta}$. Then the first exponent in (C.1) is bounded away from zero for all sufficiently large v , as are the other exponential rates. Therefore the centered discrepancy is $2^{-\Omega(v)}$. This is a sufficient criterion, not a converse or an optimal degree threshold.

C.2 Any fixed expander gap can meet the criterion

It is useful to give a deliberately loose closed-form choice. Write $\kappa = 1 - \rho$, where $0 < \kappa \leq 1$. It suffices to choose fixed integers satisfying

$$L \geq \frac{18}{\kappa}, \quad d \geq 2592 \kappa^{-5/2}. \quad (\text{C.3})$$

Then $w_\infty \geq \kappa^2/324$, so eventually $\mathbb{E}Z \geq \eta v$, where

$$\eta = \frac{\kappa^2}{648}, \quad \tau = \frac{\eta^2}{16} = \frac{\kappa^4}{6718464}.$$

The degree choice gives

$$\beta = \frac{1}{2\kappa d^2} \leq \frac{\tau}{2}.$$

Using the threshold $\eta v/2$ in the concentration argument therefore gives

$$\Pr[Z < \eta v/2 \mid E] \leq v e^{-\tau v/2} \leq e^{-\tau v/4}$$

for sufficiently large v . The same construction has at least $\eta v/4$ honoured active stages outside this mass. Consequently its centered discrepancy is at most

$$e^{-\tau v/4} + 2^{-\eta v/8} + 2^{1-v} = 2^{-\Omega(v)}. \quad (\text{C.4})$$

The implicit constants depend only on the fixed gap and chosen degree and clone count.

C.3 A fixed clique blow-up makes an arbitrary expander eligible

Theorem C.1 (Fixed-gap expander families). *Fix an integer $\Delta \geq 2$ and a real $0 < \kappa_0 \leq 1$. There is an integer M , depending only on these fixed parameters, with the following property. For every sufficiently large connected simple Δ -regular graph G satisfying*

$$\lambda_2(A_G/\Delta) \leq 1 - \kappa_0,$$

the host $H = G[K_M]$ has fixed degree and admits constants $a, b > 0$, independent of $|V(G)|$, such that

$$\text{monSIZE}(F_H \pm \epsilon P_H) \geq 2^{bN} \quad (2^{-aN} \leq \epsilon \leq 1/2), \quad (\text{C.5})$$

where $N = (|V(H)| - 1) \deg(H)$ is its actual directed-arc variable count. The circuit model, total communication predicate, and coefficient definitions are exactly those of the Theorem 2.1. If G is a uniform explicit family, the determinant upper bound is uniform as well, with the same convention on the epsilon parameter.

Proof. Put

$$\kappa = \frac{\Delta \kappa_0}{\Delta + 1}.$$

Choose a fixed integer s so large that

$$d = s(\Delta + 1) - 1 \geq 2592 \kappa^{-5/2},$$

and put $B = G[K_s]$. This is connected, simple, and d -regular. Its adjacency matrix is

$$A_B = (A_G + I) \otimes J_s - I.$$

The subspace constant within each s -clique has normalized eigenvalues

$$\frac{s(\Delta\theta + 1) - 1}{d},$$

where θ runs over the normalized eigenvalues of G . The orthogonal subspace has normalized eigenvalue $-1/d$. For every nonconstant lifted eigenvector,

$$\frac{s(\Delta\theta + 1) - 1}{d} \leq 1 - \frac{s\Delta\kappa_0}{d} \leq 1 - \kappa.$$

Also $-1/d \leq 1 - \kappa$, since $0 < \kappa < 1$. Thus B satisfies the second-eigenvalue hypothesis with $\rho = 1 - \kappa$.

Choose $L \geq 18/\kappa$. The preceding estimates provide centered discrepancy 2^{-cv} for some fixed $c > 0$, where $v = d|V(B)| = ds|V(G)|$, on

$$H = B[K_{dL}] \cong G[K_{sdL}].$$

Set $M = sdL$. The blow-up factor is a fixed integer. With $K = L((d+1)dL - 1)$, the actual count satisfies $N < Kv$. The shared-gate and coefficient transfer gives $s_{\text{circ}} \geq \epsilon/(3\gamma)$. For sufficiently large graphs, take $a = c/(2K)$, $b = c/(3K)$, and absorb the constant three exactly as in the Theorem 2.1 to obtain (C.5).

The graph transformation is a fixed local blow-up. Its polynomial and determinant are uniform whenever the original graph family is uniform. $\square$

The same two-level coefficient and sharp uniform approximation corollaries hold. This theorem does not assert hardness on the original G without the clique blow-up, nor constants uniform in a gap that shrinks with the graph size.

D Finite verification and reproducibility

The proofs above do not depend on experimental extrapolation. Dedicated computations check the counting conventions, local gadgets, and circuit transformations on finite examples. The parity identity (3.9) is proved in Section 3.2 for every admissible layout and every bit assignment; the gadget checks below test its implementation and are not premises of that proof. Table 1 summarizes the principal checks. The accompanying ancillary files contain the Python implementations, exact result records, and a manifest of the files used.

The counting check includes a circuit of formal degree $4 \cdot 2^{256}$, whose normalized output has 16 monomials. The shared-gate check allows multiple productive uses of the selected gate, and tests the global substitution at all uses. Incorrect first-exit samplers, root-exit modifications, single-use charges, and omitted variable-restoration steps were rejected. The additional charge check distinguishes different DAG nodes that compute the same polynomial and verifies the level-set decomposition with exact rational arithmetic. These finite checks supplement Lemma 4.1 and Proposition 4.2.

The toy graphs do not meet the large-degree and spectral hypotheses of Theorem 2.1. These computations check finite identities and implementations, not the universal spectral estimates, concentration inequalities, or asymptotic circuit lower bound. Reproducibility of a finite check is also distinct from independent mathematical review.

Check	Finite coverage
BEST counting and sampling	47,976 local covers and 16,334 last-exit descriptions on $C_3, C_4, K_4, K_{3,3}$; exact multiplicities and matrix-tree counts
Permutation exposure	666,600 exact conditional-mean configurations
Direct parity gadget	26,624 bit inputs on 46 layouts; all arcs checked in the physical host
Smaller-host implementation	48 layouts, 12,288 bit inputs, 1,728 locality checks, and eight exact state-spectrum identities
Pair-margin estimate	2,016 cluster colorings and 93,610 discrete density pairs
Balanced shared-gate removal	2,468 products from 120 DAGs; 6,588 exact output coefficients
Shared-gate coefficient charge	10 global-replacement fixtures, including 2^{256} path multiplicity and reciprocal 2^{512} rescaling; 20,480 rectangles for 80 signed measures
Least-degree component	12,800 gate-component comparisons in 400 DAGs
Counting normalization	20,789 gate profiles and exact polynomials in 400 DAGs; 10,240 Boolean input points

Table 1: Finite checks completed for the source arguments.

References

- [1] Arkadev Chattopadhyay, Rajit Datta, and Partha Mukhopadhyay. Lower bounds for monotone arithmetic circuits via communication complexity. In *Proceedings of STOC 2021*, pages 786–799, 2021. [doi:10.1145/3406325.3451069](https://doi.org/10.1145/3406325.3451069).
- [2] Arkadev Chattopadhyay, Rajit Datta, Utsab Ghosal, and Partha Mukhopadhyay. Monotone complexity of spanning tree polynomial re-visited. In *13th Innovations in Theoretical Computer Science Conference*, volume 215 of *LIPIcs*, pages 39:1–39:21, 2022. [doi:10.4230/LIPIcs.ITCS.2022.39](https://doi.org/10.4230/LIPIcs.ITCS.2022.39).
- [3] Arkadev Chattopadhyay, Utsab Ghosal, and Partha Mukhopadhyay. Robustly separating the arithmetic monotone hierarchy via graph inner-product. In *42nd IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science*, volume 250 of *LIPIcs*, pages 12:1–12:20, 2022. [doi:10.4230/LIPIcs.FSTTCS.2022.12](https://doi.org/10.4230/LIPIcs.FSTTCS.2022.12).
- [4] Matthew Farrell and Lionel Levine. Multi-Eulerian tours of directed graphs. Preprint, 2015. [arXiv:1509.06237](https://arxiv.org/abs/1509.06237).
- [5] Sergey Fomin, Dima Grigoriev, and Gleb Koshevoy. Subtraction-free complexity, cluster transformations, and spanning trees. Max-Planck-Institut für Mathematik, Preprint 2013 (36), 2013. https://webdoc.sub.gwdg.de/ebook/serien/e/mpi_mathematik/2013/36.pdf.
- [6] Pavel Hrubeš. On ϵ -sensitive monotone computations. *Computational Complexity*, 29, article 6, 2020. [doi:10.1007/s00037-020-00196-6](https://doi.org/10.1007/s00037-020-00196-6).
- [7] Mark Jerrum and Marc Snir. Some exact complexity results for straight-line computations over semirings. *Journal of the ACM*, 29(3):874–897, 1982. [doi:10.1145/322326.322341](https://doi.org/10.1145/322326.322341).
- [8] Stasys Jukna. Lower bounds for monotone counting circuits. *Discrete Applied Mathematics*, 213:139–152, 2016. [doi:10.1016/j.dam.2016.04.024](https://doi.org/10.1016/j.dam.2016.04.024).
- [9] Alexander Lubotzky, Ralph Phillips, and Peter Sarnak. Ramanujan graphs. *Combinatorica*, 8(3):261–277, 1988. <https://math.huji.ac.il/~alexclub/PAPERS/ramanujan%20graphs/ramanujanGraphs.pdf>.

- [10] James Martens and Venkatesh Medabalimi. On the expressive efficiency of sum product networks. Preprint, version 3, 2015. [arXiv:1411.7717v3](https://arxiv.org/abs/1411.7717v3).
- [11] Robert Peharz, Sebastian Tschiatschek, Franz Pernkopf, and Pedro Domingos. On theoretical properties of sum-product networks. In *Proceedings of the Eighteenth International Conference on Artificial Intelligence and Statistics*, volume 38 of *PMLR*, pages 744–752, 2015. <https://proceedings.mlr.press/v38/peharz15.html>.
- [12] Jesse Thorner and Asif Zaman. Refinements to the prime number theorem for arithmetic progressions. *Mathematische Zeitschrift*, 306(3), article 54, 2024. [doi:10.1007/s00209-023-03414-3](https://doi.org/10.1007/s00209-023-03414-3).